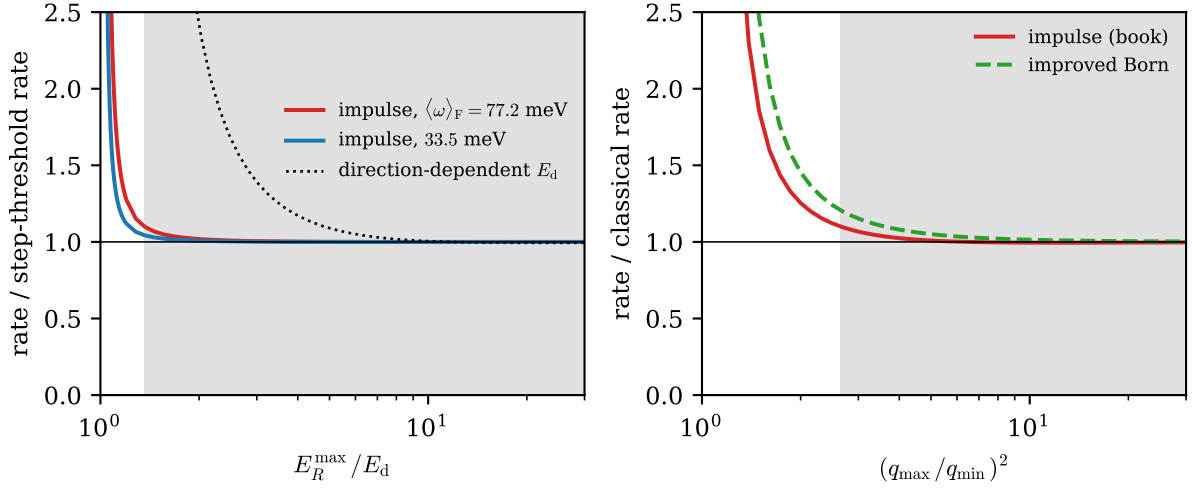
(a) F recoil in LiF
rate / step-threshold rate
2.5
2.0
1.5
1.0
0.5
0.0
10^0
10^1
$E_R^{\max}/E_{\mathrm{d}}$
impulse, $\langle\omega\rangle_{\mathrm{F}} = 77.2$ meV
impulse, 33.5 meV
direction-dependent E_{d}
(b) H_2 bond
rate / classical rate
impulse (book)
improved Born
$(q_{\max}/q_{\min})^2$